

Social Interactions and the Institutions of Local Government

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Many economic processes are influenced by externalities within groups. Educational outcomes depend on peer-group interactions between students, which may help explain the persistence of income inequality and the stability of subcultures and social classes.¹ Crime rates exhibit a geographic pattern that strongly suggests the presence of interactions between potential criminals. There is also substantial evidence that amenities are influenced by interactions between neighbors, and that interactions between firms influence labor productivity.²

This paper considers how social interactions affect the institutions of local government. Specifically, we show how social interactions encourage consumers to withdraw from the traditional public sector and join exclusive groups that regulate the activities of their members. Examples include familiar organizations like exclusive suburbs and private schools and new or newly popular institutions like private governments and charter schools. Each of these institutions mediates social interactions by excluding some agents and altering the actions of others. We view the formation of these institutions as a kind of secession, since members withdraw from the civic whole and limit their interactions to other group members.

These new organizations are increasingly important, surprisingly powerful, and highly con-

troversial. One of the most widespread innovations in local government in recent years has been the rise of residential private government, including common interest developments (CIDs) and homeowner associations (HOAs). Evan McKenzie (1996) reports that the number of CIDs in the United States grew from a few hundred in the 1960's to 150,000 in 1993, and that their populations now total at least 32 million people. CIDs and HOAs are generally formed by real estate developers, and are eventually governed by an elected board of members. CIDs limit interactions with the rest of the world in a number of ways, most notoriously by building walls (Edward J. Blakely and Mary Gail Snyder, 1997). They tax their members to pay for the local public services they provide (primarily street maintenance, trash collection, and policing), collectively own and manage shared facilities (recreation centers, parks, and sometimes streets), and regulate both property use and individual conduct through covenants, conditions, and restrictions (CCRs) established by the developer. The regulatory activities of CIDs are impressive. Activities that have been prohibited include "flying the flag, delivering newspapers, parking pickup trucks in the driveway, kissing outside the front door, using one's own back door too much, building fences, painting the exterior certain colors, having pets, working from one's home, marrying people below a certain age, and even having children" (McKenzie, 1996 p. 4). In spite of, or perhaps because of, these regulations, CIDs provide a higher level of amenities than is available in public developments. However, critics view them as undemocratic and discriminatory private governments operating outside the constitutional restrictions that public governments face.

A primary goal of this paper is to provide a model that captures the common and general features of the new institutions of local government. To that end, we develop a model of local secession motivated by social interactions and supported by regulation. The model has three essential elements. First, heterogeneous agents belong to groups, and each takes an action that

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¹ See Anita A. Summers and Barbara L. Wolfe (1977), J. Vernon Henderson et al. (1978), Roland Bénabou (1993, 1996), Steven N. Durlauf (1996), and George A. Akerlof (1997).

² See Joseph Gyourko and Joseph Tracy (1991), Raaj Sah (1991), William N. Evans et al. (1992), Charles F. Manski (1993), Edward L. Glaeser et al. (1996), and John M. Quigley (1998).

impacts the welfare of other group members. Thus, social interactions arise from individual contributions to group quality, a collective good. Second, each agent makes a membership or secession decision. There are two groups, an exclusive group and the remainder. The extent of secession (the size of the exclusive group) is variable and endogenous. Third, the secessionist group regulates the activities of its members. This directly increases the quality of the exclusive group. Since those who are unwilling to abide by the regulation choose not to secede, regulation also serves as an exclusion device.

The model generates a number of interesting results. First, while secession increases the quality of the exclusive group, it decreases the quality of the remainder. Second, some secession may be efficient, but there is too much secession in equilibrium. Third, a planner who lacked the ability to restrict secession directly could still enhance efficiency by requiring more stringent regulation. Thus, far from being undesirable, the many regulations imposed by exclusive organizations may be useful second-best instruments. Fourth, the objectives pursued by the exclusive group have important effects on resource allocation. For example, a profit-maximizing exclusive group might choose to regulate less stringently than one that pursued welfare maximization.

The paper is organized as follows. Section I presents the model, Section II characterizes equilibrium, and Section III solves for the efficient allocation. Section IV considers entrepreneurship and voting. Section V concludes.

I. The Model

A. Social Interactions

Each agent takes an action that influences the quality of his or her group. We model this social interaction as follows. Let $a(\theta, A_G)$ represent the action of a type θ agent in a group of quality A_G . Define the quality of group G by,

$$(1) \quad A_G = \frac{\int_{\theta \in G} a(\theta, A_G) h(\theta) d\theta}{\int_{\theta \in G} h(\theta) d\theta},$$

where $h(\theta)$ is the density of types with support $[\theta, \bar{\theta}]$, and the mass of agents is one. In this simple formulation, group quality equals the mean individual action. Note that group quality is endogenous in our model in the sense that it arises from the actions of agents.³

The utility of a type θ agent taking action a in a group of quality A_G is

$$(2) \quad U(\theta, a, A_G) = y + u(\theta, a, A_G) - ka,$$

where y is income and k is the cost of individual action. We assume that the subutility function $u(-)$ is continuously differentiable and satisfies $u_a > 0$, $u_A > 0$, $u_{a\theta} > 0$, $u_{A\theta} > 0$, and $u_{aa} < 0$. Thus, the marginal utility of both individual action and group quality increase with type.

Utility maximization implies

$$(3) \quad U_a \equiv u_a - k = 0.$$

Equation (3) implicitly defines the utility-maximizing action $a(\theta, A_G)$ and the type for which action a is maximizing $\theta(a, A_G)$. We assume $u_a(\theta, 0, A_G) > k$ and $\lim_{a \rightarrow \infty} u_a(\theta, a, A_G) < k$ for all feasible A_G , so that $a(\theta, A_G)$ is positive and finite. Note that the utility-maximizing action is increasing in θ : $a_\theta = -U_{a\theta}/U_{aa} > 0$. The utility-maximizing action may be increasing or decreasing in A_G , depending on whether individual action and group quality are complements ($U_{aA} > 0$) or substitutes ($U_{aA} < 0$): $a_A = -U_{aA}/U_{aa}$.

B. Secession

We model secession as the bifurcation of the population into two groups, a high group (group H) consisting of all agents above some

³ This contrasts with the tradition of supposing that peer-group effects arise from the exogenous characteristics of agents (for example, Richard J. Arnott and John Rowse, 1987; Charles A. M. de Bartolome, 1990; Dennis Epple and Richard E. Romano, 1998; Thomas J. Nechyba, 1998), and is crucial to the regulatory nature of secessionist institutions. An exception is Bénabou (1993), who examines a model in which individuals choose a skill level that impacts the cost of education for others in a city.

critical type θ^* , and a low group (group L) consisting of the remainder.⁴ Agents in the high group do not contribute to the quality of the low group, and vice versa. We require that secession be incentive compatible—agents join the high group only when it is in their interest to do so.

C. Exclusion and Regulation

Secession requires exclusion. There are a number of ways that agents could be excluded from the high group. For example, one could imagine that membership is controlled by a manager with the ability to exclude directly. In this case, the manager would presumably choose θ^* to maximize an objective function subject to the incentive-compatibility requirement discussed above. One obvious difficulty with direct exclusion is that it requires that θ be observable to the manager. In our case, θ indexes an agent's preference for individual action and group quality, which is almost certainly unobservable. One could also imagine a model where a membership price serves to exclude some agents from the high group.⁵

In what follows, we show that when social interactions arise from individual contributions to a collective good, exclusion can be accomplished through the regulation of individual actions. Specifically, we assume that the high group sets a minimum action $\underline{q}$ that is required of all members. When this minimum action is relatively large, the cost of adhering to the regulation causes some low-type agents to choose not to join the exclusive group. In this way, regulation serves as an exclusion device. Further, regulation dominates pricing

in this setting because regulation, in contrast to both pricing and direct exclusion, can increase individual actions and improve group quality. Thus, regulation not only excludes, it also enhances social interactions within the high group. It is worth observing that our regulatory policy does not require that θ or $a(\theta, A_G)$ be contractible. All that is required is that the regulator can verify compliance with the minimum standard.⁶

The impact of regulation on the quality of the high group depends on the number of agents for whom the minimum standard is binding. Agents in the high group take action $a(\theta, A_H)$ or $\underline{q}$, whichever is larger. If $\underline{q} < a(\bar{\theta}, A_H)$, then types above $\theta(\underline{q}, A_H)$, where $a(\theta(\underline{q}, A_H), A_H) = \underline{q}$, are not affected directly by the minimum standard. Types below $\theta(\underline{q}, A_H)$ take the minimum action $\underline{q}$. We say that regulation is weak in this case. If $\underline{q} \geq a(\bar{\theta}, A_H)$, then all agents in the high group take action $\underline{q}$ and we define $\theta(\underline{q}, A_H) = \bar{\theta}$. We say that regulation is strong in this case.

For $\theta^* \in (\underline{\theta}, \bar{\theta})$, the mappings that determine the qualities of the low and high groups are:

$$(4) \quad A_L = \Phi_L(A_L)$$

$$\equiv \frac{\int_{\underline{\theta}}^{\theta^*} a(\theta, A_L) h(\theta) d\theta}{H(\theta^*)},$$

and

$$(5) \quad A_H = \Phi_H(A_H)$$

$$\equiv \frac{\int_{\theta^*}^{\theta(\underline{q}, A_H)} \underline{q} h(\theta) d\theta + \int_{\theta(\underline{q}, A_H)}^{\bar{\theta}} a(\theta, A_H) h(\theta) d\theta}{1 - H(\theta^*)}.$$

We assume that (4) and (5) give unique and positive solutions, $A_L(\theta^*)$ and $A_H(\theta^*, \underline{q})$.⁷ The comparative statics of A_L and A_H are:

⁴ The fixed costs of secession presumably limit the number of groups. However, the equilibrium number of secessionist groups is not analyzed here. The assumption that there is only one exclusive group is extremely convenient, but is not essential. Our key results would continue to hold if there were many exclusive groups. The essential assumption is that there is a remainder of agents who are left behind by secession.

⁵ Helsley and Strange (1998) consider a related model of private government in which a group of dissatisfied agents provides a collective good supplement to its members. In that model, pricing and public spending serve as exclusion mechanisms.

⁶ Since θ is unobservable, there can be no type-prices of the kind employed in Bryan Ellickson et al. (1997) or John P. Conley and Myrna Wooders (1999).

⁷ $|a_A(\theta, A_G)| < 1$ ensures that (4) and (5) have unique solutions. $a(\theta, A_G) > 0$ ensures that A_L and A_H are positive. Details are available on request.

$$(6) \quad \frac{\partial A_L}{\partial \theta^*} = \frac{\frac{h(\theta^*)}{H(\theta^*)} (a(\theta^*, A_L) - A_L)}{1 - \frac{\int_{\underline{\theta}}^{\theta^*} a_A(\theta, A_L) h(\theta) d\theta}{H(\theta^*)}}.$$

$$(7) \quad \frac{\partial A_H}{\partial \theta^*} = \frac{\frac{h(\theta^*)}{1 - H(\theta^*)} (A_H - \underline{a})}{1 - \frac{\int_{\theta(\underline{a}, A_H)}^{\bar{\theta}} a_A(\theta, A_H) h(\theta) d\theta}{1 - H(\theta^*)}},$$

$$\frac{\partial A_H}{\partial \underline{a}} = \frac{\frac{1}{1 - H(\theta^*)} (H(\theta(\underline{a}, A_H)) - H(\theta^*))}{1 - \frac{\int_{\theta(\underline{a}, A_H)}^{\bar{\theta}} a_A(\theta, A_H) h(\theta) d\theta}{1 - H(\theta^*)}}.$$

Assuming that the denominators in (6) and (7) are positive, $\partial A_L / \partial \theta^* > 0$, $\partial A_H / \partial \theta^* \geq 0$, and $\partial A_H / \partial \underline{a} > 0$.⁸ Further, if regulation is strong, then $\theta(\underline{a}, A_H) \equiv \bar{\theta}$, $A_H = \underline{a}$, and so $\partial A_H / \partial \underline{a} = 1$ and $\partial A_H / \partial \theta^* = 0$. Thus, the quality of both groups increases with θ^* , and the quality of the high group increases with the level of the minimum standard.

While this model does not capture every characteristic of the new institutions of local government, it captures three important ones: there are social interactions, there is secession, and there is regulation of members.⁹ We believe that this model applies to many innovations in local government, including CIDs, exclusive suburbs, downtown associations, and charter or magnet schools. That social

interactions and secession play important roles in each of these institutions is difficult to dispute.

It is also clear that regulation plays a role. CIDs and exclusive suburbs are notorious for their regulatory vigor (William A. Fischel, 1985; McKenzie, 1994). Innovative schools may have performance standards or require parental participation and purchases of materials (i.e., computers or school uniforms). Downtown associations may impose design standards or mandate participation in community-based policing. Regulation is probably not as important for charter schools or downtown associations as for CIDs or exclusive suburbs. One can speculate on the reasons for this. It may be that the peer-group effect arises from individual characteristics (e.g., smart children) rather than actions. Or it may be difficult to ensure that the regulated actions (e.g., ensuring that the children watch less television) take place. Nonetheless, these institutions do carry out some regulation, and there is reason to believe that in the future they may be more regulatory. The closest cousin of a downtown association is a shopping mall, while private schools are the nearest relatives of charter or magnet schools. Both of these institutions regulate aggressively.

D. Equilibrium

A solution to the model is a Nash equilibrium of the following simultaneous move game. Each agent secedes if it is in her interest to do so, determining θ^* . Regulation is chosen to satisfy the objectives of the high group (we consider several), determining $\underline{a}$. Agents choose individual actions as discussed above, determining $a(\theta, A_G)$. In equilibrium, all these choices are mutually consistent.

II. Equilibrium with Welfare Maximization

This section characterizes equilibrium when regulation is chosen to maximize the welfare of the high group. This equilibrium embodies a public interest view of social choice: even if there is rent-seeking, regulators seek to maximize the rents that they will divide (Roger G. Noll, 1989).

⁸ $|a_A(\theta, A_G)| < 1$ ensures that is true.

⁹ We have deliberately ignored the expenditures of the new institutions. Charles M. Tiebout (1956), Wallace E. Oates (1981), David E. Wildasin (1986), Suzanne Scotchmer and M. Wooders (1987), Jan K. Brueckner and Kangoh Lee (1989), Helsley and Strange (1991, 1998), Raquel Fernández and Richard Rogerson (1996), and Todd Sandler and John Tschirhart (1997) are only a few of the many papers on this topic.

A. Regulation

The regulator chooses $\underline{q}$ to maximize high-group welfare taking the secession margin θ^* as fixed. The welfare of the high group is

(8)

$$W_H = \int_{\theta^*}^{\theta(\underline{q}, A_H)} U(\theta, \underline{q}, A_H) h(\theta) d\theta + \int_{\theta(\underline{q}, A_H)}^{\bar{\theta}} U(\theta, a(\theta, A_H), A_H) h(\theta) d\theta.$$

This additive welfare function makes use of our transferable utility assumption (there are no income effects).

Using (3), the first-order conditions for a maximum of W_H are

$$(9) \quad \frac{\partial W_H}{\partial \underline{q}} = \int_{\theta^*}^{\theta(\underline{q}, A_H)} U_a(\theta, \underline{q}, A_H) h(\theta) d\theta + \left\{ \int_{\theta^*}^{\theta(\underline{q}, A_H)} U_A(\theta, \underline{q}, A_H) \frac{\partial A_H}{\partial \underline{q}} h(\theta) d\theta + \int_{\theta(\underline{q}, A_H)}^{\bar{\theta}} U_A(\theta, a(\theta, A_H), A_H) \times \frac{\partial A_H}{\partial \underline{q}} h(\theta) d\theta \right\} = 0.$$

As noted above, when regulation is strong $\theta(\underline{q}, A_H) \equiv \bar{\theta}$ and $\partial A_H / \partial \underline{q} = 1$. Regulation has two effects on the welfare of the high group. First, it causes some members to contribute more than they would like. The first term in (9) is the marginal decrease in the utility of regulated members as a result of this distortion. This term is negative since $U_a < 0$ whenever the regulation is binding by (3). Second, regulation increases the mean individual action and thus the quality of the high group. The second (bracketed) term in (9) gives the marginal increase in the utility of all members

as a result of this externality. This term is positive since $\partial A_H / \partial \underline{q} > 0$ from (7). At the maximizing regulation, these marginal effects are offsetting. Assuming that W_H is strictly concave, (9) implicitly defines the welfare-maximizing regulation as a continuous function of θ^* : $\underline{q}(\theta^*)$.

Evaluating (9) at $\underline{q} = \underline{q}(\theta^*, A_H)$, we have

$$(10) \quad \frac{\partial W_H}{\partial \underline{q}} \text{ at } \underline{q} = \underline{q}(\theta^*, A_H) = \int_{\theta^*}^{\bar{\theta}} U_A(\theta, a(\theta, A_H), A_H) \times \frac{\partial A_H}{\partial \underline{q}} h(\theta) d\theta > 0.$$

Thus, regulation is binding on the marginal member of the high group. Regulation will be strong (binding on all members of the high group) if

$$(11) \quad \frac{\partial W_H}{\partial \underline{q}} \text{ at } \underline{q} = \underline{q}(\bar{\theta}, A_H) = \int_{\theta^*}^{\bar{\theta}} (U_a(\theta, a(\bar{\theta}, A_H), A_H) + U_A(\theta, a(\bar{\theta}, A_H), A_H)) h(\theta) d\theta > 0.$$

Strong regulation is more likely the larger the contribution of group quality to utility, and the smaller the cost of individual action.

B. Secession

A type θ agent secedes if $U(\theta, \text{Max}\{\underline{q}, a(\theta, A_H)\}, A_H) - U(\theta, a(\theta, A_L), A_L) \geq 0$. Let $p(\theta, \underline{q}, A_H, A_L)$ represent willingness to pay to join the high group. Since regulation is binding on the marginal member of the high group, $\text{Max}\{\underline{q}, a(\theta^*, A_H)\} = \underline{q}$, and we have

$$(12) \quad p(\theta^*, \underline{q}, A_H, A_L) \equiv U(\theta^*, \underline{q}, A_H) - U(\theta^*, a(\theta^*, A_L), A_L) = 0.$$

The following result establishes the nature of secession.

PROPOSITION 1 [Secession]: *Suppose $A_H > A_L$ and that there is a $\theta^* \in (\underline{\theta}, \bar{\theta})$ such that $p(\theta^*, \underline{a}, A_H, A_L) = 0$. Then all agents of type $\theta > \theta^*$ will choose to join and all those of type $\theta < \theta^*$ will choose not to join.*¹⁰

The essence of the proof is that agents above the margin have higher utility as members holding group quality fixed since $U_{\theta a} > 0$ (regulation is less onerous) and $U_{\theta A} > 0$ (group quality is more valuable). We assume that the equilibrium is locally stable. This requires $dp(\theta^*, \underline{a}, A_H(\theta^*), A_L(\theta^*))/d\theta^* > 0$. We will now show that such an equilibrium exists.

C. Equilibrium

An equilibrium of the game satisfies (4), (5), (9), and (12), or equivalently is a solution of $P(\theta^*) \equiv p(\theta^*, \underline{a}(\theta^*), A_H(\theta^*), A_L(\theta^*)) = 0$.

PROPOSITION 2 [Existence]: *Suppose $P(\underline{\theta}) < 0$. Then there is at least one interior stable equilibrium.*

An equilibrium exists because the lowest types never secede, the highest types always secede, and $P(\theta^*)$ is a continuous function. In what follows, we assume that $dP(\theta^*)/d\theta^* > 0 \forall \theta \in (\underline{\theta}, \bar{\theta})$, so that the equilibrium is unique. We present an example below where this condition holds.

D. Group Quality and Welfare

Prior to secession, all agents are member of the universal group. The quality of the universal group is given by

$$(13) \quad A_0 = \int_{\underline{\theta}}^{\bar{\theta}} a(\theta, A_0) h(\theta) d\theta.$$

The effect of secession on group quality is summarized by the following.

PROPOSITION 3 [Equilibrium Group Qualities]: *At an interior equilibrium, the quality of the high group exceeds the quality of the universal group, which in turn exceeds the quality of the low group.*

When regulation is such that some agents find that it is not in their interest to secede, high-group quality improves because low-type agents are effectively excluded and because regulation is binding on some or all members of the high group. This result would hold even if there were many secessionist groups, so long as there is a remainder of agents who are excluded. The following is a corollary.

PROPOSITION 4 [Cream Skimming]: *All members of the low group are made worse off by secession.*

Thus, it is not surprising that secession of the form modeled here is controversial. Every agent who is left behind is made worse off by the decline in group quality.

The effect of secession on the welfare of the high group is more complex. The change in the utility of a high-group member is $U(\theta, \text{Max}\{a, a(\theta, A_H)\}, A_H) - U(\theta, a(\theta, A_0), A_0)$. There are several cases. First, all unregulated members of the high group are better off with secession, since $U_A > 0$ and $A_H - A_0 > 0$ (Proposition 3). Second, whether or not regulation is strong, since regulation maximizes high-group welfare, the highest-type members are better off. Third, the marginal member is as well off joining as not. Since all nonmembers are worse off, so is the marginal member. Thus, although the marginal member benefits from a higher-quality group, the regulation is sufficiently stringent to outweigh this. By continuity, members with only slightly stronger preferences than the marginal member are also worse off. Note that if there were many secessionist groups, the members of the least selective would be worse off by the same argument. To summarize, some members are worse off, while others are better off.

E. Example

We will now illustrate the equilibrium with a simple computational model.¹¹ Suppose $u(\theta, a,$

¹⁰ All proofs are collected the Appendix.

¹¹ Details are available on request.

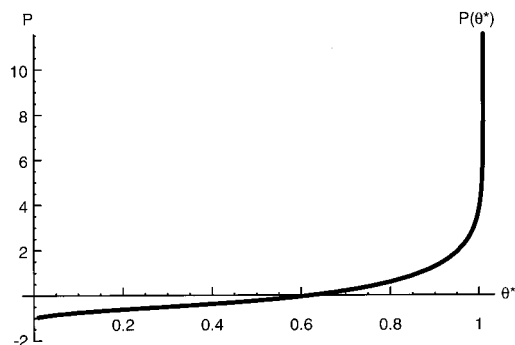


FIGURE 1. EQUILIBRIUM SECESSION

Note: See the notes to Table 1.

$A_G) = \theta(10 + \ln(a) + vA_G)$, and that θ is uniformly distributed on $[\underline{\theta}, \bar{\theta}]$. Suppose also that regulation is strong, a supposition that will be confirmed shortly. Then the welfare-maximizing regulation [from (9)] is given by

$$(14) \quad \underline{a} = \frac{\bar{\theta} + \theta^*}{2k - v(\bar{\theta} + \theta^*)},$$

and the willingness to pay of the marginal member of the high group [from (13)] is

$$(15) \quad p(\theta^*, \underline{a}) \\ = \theta^* \left(1 + \ln \left(\frac{\underline{a}k}{\theta^*} \right) \right) + \underline{a}(v\theta^* - k) \\ - \frac{v\theta^*}{2k} (\bar{\theta} + \theta^*).$$

Composing the former into the latter generates the $P(\theta^*)$ function shown in Figure 1, where we assume $v = 1$, $k = 1.01$, and $[\underline{\theta}, \bar{\theta}] = [0.01, 1.01]$.¹² Note that $P(\theta^*)$ is strictly increasing and has a unique interior root in this example. As detailed in Table 1, the equilibrium value of $\theta^* = 0.626$, so around 37 percent of the population joins the high group. Because utility is

¹² With these parameters, by (11), the critical value of θ^* above which regulation is strong is zero.

TABLE 1—A COMPUTATIONAL MODEL

	Welfare-maximizing equilibrium	Efficient allocation	Voting equilibrium	Entrepreneurial equilibrium
θ^*	0.626	0.671	0.626	0.807
$a(\theta^*)$	0.620	0.664	0.620	0.799
$\underline{a}$	4.262	4.950	4.262	3.982
A_0	0.505	0.505	0.505	0.505
A_L	0.315	0.337	0.315	0.405
A_H	4.262	4.950	4.262	3.982

Notes: This table assumes $u(\underline{a}) = 10 + \ln \underline{a} + vA$, $v = 1$, $H(\theta) = \theta/(\bar{\theta} - \underline{\theta})$, $[\underline{\theta}, \bar{\theta}] = [0.01, 1.01]$, and $k = 1.01$. All equilibria involve strong regulation.

separable, the individual contribution is independent of group quality. Thus, we have $a(\theta^*, A_L) = a(\theta^*, A_H) \equiv a(\theta^*) = 0.620$, while $\underline{a} = 4.262$. The equilibrium group qualities are $A_0 = 0.505$, $A_L = 0.315$, and $A_H = \underline{a} = 4.262$.

III. Efficiency

This section considers efficient secession and regulation. First, we compare the equilibrium to a constrained-efficient allocation where the planner can control secession and regulation but not individual contributions. Second, we consider efficiency when the planner can control only regulation.

A. A Constrained-Efficient Allocation

In this case, the planner chooses $\underline{a}$ and θ^* to maximize aggregate welfare, $W = W_L + W_H$, where W_H is defined in (8) and

$$(16) \quad W_L = \int_{\underline{\theta}}^{\theta^*} U(\theta, a(\theta, A_L), A_L) h(\theta) d\theta.$$

We suppose that the planner cannot directly set the contribution of an individual agent. This means that the planner can influence group quality only by regulating and by partitioning the agents into two groups.

The first-order condition for efficient regulation is $\partial W / \partial \underline{a} = 0$. Since W_L does not depend on $\underline{a}$ directly, $\partial W / \partial \underline{a} = 0$ implies $\partial W_H / \partial \underline{a} = 0$, and the first-order condition for efficient regulation is identical to (9), the first-order condi-

tion for equilibrium regulation. Thus, given θ^* , equilibrium regulation is efficient.

The first-order condition for efficient secession implies that at an interior solution

$$\begin{aligned}
 (17) \quad & p(\theta^*, \underline{a}, A_H(\theta^*, \underline{a}), A_L(\theta^*)) \\
 &= \int_{\theta}^{\theta^*} U_A(\theta, a(\theta^*, A_L), A_L) \\
 &\quad \times \frac{\partial A_L}{\partial \theta^*} h(\theta) d\theta \\
 &\quad + \int_{\theta^*}^{\bar{\theta}} U_A(\theta, \max\{\underline{a}, A_H\}, A_H) \\
 &\quad \times \frac{\partial A_H}{\partial \theta^*} h(\theta) d\theta.
 \end{aligned}$$

If regulation is strong, then $\partial A_H / \partial \theta^* = 0$ and the second term in (17) vanishes. One interesting implication of (17) is that when social interactions arise because of differences in individual tastes, it is possible that some secession may be efficient. This idea is summarily dismissed in some of the popular analysis of secession (e.g., Robert Reich, 1991) and it is obscured in models where the extent of secession is fixed (e.g., de Bartolome [1990] and Bénabou [1993, 1994]).

The most important implication of (17) is the following.

PROPOSITION 5 [Excessive Secession]: *The constrained-efficient allocation has less secession than the interior Nash equilibrium.*

Proposition 5 is a consequence of the externalities associated with secession. When regulation is weak, any increase in the size of the high group decreases the quality of both groups. When regulation is strong, only the quality of the low group is affected. In either case, since the marginal individual ignores these external costs when deciding whether or not to secede, the equilibrium involves more secession than the constrained-efficient allo-

cation.¹³ If social interactions are very important (large U_A), then this problem becomes more serious. Proposition 5 is driven by the presence of a remainder of excluded agents who are made worse off by secession. The degree of secession would be excessive even if there were many exclusive groups. The second column of Table 1 shows that some secession (roughly 33 percent of the population) is efficient in our example, and that there is too much secession in the equilibrium with welfare maximization.

An interesting special case occurs when $u_a = 0$. In this case, the actions of agents confer no individual benefits, and group quality is more like a pure public good. In the low group, $a(\theta, A_L) = 0$, and so $\partial A_L / \partial \theta^* = 0$. Obviously, the high group will regulate strongly, and so $\partial A_H / \partial \theta^* = 0$. Substituting these into (17) shows that efficient secession requires $p(\theta^*, \underline{a}, A_H(\theta^*, \underline{a}), A_L(\theta^*)) = 0$. The equilibrium is efficient in this case because adding a marginal member to the high group does not impact either high- or low-group quality. Of course, the actions with which this paper is concerned—contributions to a child's education, a neighborhood's safety or charm, and a business district's commercial attractiveness—almost certainly have $u_a > 0$.

B. Constrained Efficiency with Equal Utility

Equation (17) implies that in the constrained-efficient allocation, the marginal member of the high group has greater utility than the marginal nonmember. To support this allocation, the planner must be able to control the degree of secession directly. If instead the planner must respect the incentive-compatible membership choices of agents, then his objective is to maximize welfare subject to equal utility at the margin. The planner's problem in this case is to

¹³ Ellickson et al. (1997) establish efficiency—literally, core equivalence—in a model of nonanonymous crowding based on agents' exogenous types. Conley and Wooders (1999) prove a similar result to a model where exogenous "genetic" types make choices that determine their endogenous "crowding" types. In both cases, efficiency requires the ability to charge type prices. In our model, charging such prices is impossible because individual choices of action are not contractible beyond setting a minimum standard.

choose $\underline{q}$ and θ^* to maximize W subject to $p(\theta^*, \underline{q}, A_H(\theta^*, \underline{q}), A_L(\theta^*)) = 0$.

PROPOSITION 6 [Second-Best Efficiency]:

A planner who cannot control secession directly can increase welfare by regulating more strictly.

The key feature of the proof is that a more stringent regulation discourages secession, improving efficiency when secession cannot be controlled directly.

This kind of policy is a realistic possibility in many of the situations that we have been discussing. Charter schools can be required to impose stringent standards. CIDs and other private governments can be encouraged to regulate vigorously. But these are not the kinds of policies that have been adopted or are currently being considered. If anything, innovations in government that have led to increased regulation have been met with hostility.

IV. Alternate Objectives

A. Entrepreneurial Equilibrium

There are increasingly many instances of secession where the active parties are entrepreneurs. CIDs (and their regulations) are usually established by land developers, and these developers typically control the community until it is completed and largely occupied. Several school districts in large U.S. cities have recently experimented with the provision of primary education by profit-maximizing corporations.

Suppose that an entrepreneurial agent chooses a level of regulation $\underline{q}$ and a price p for admission to the high group to maximize profit. This is equivalent to supposing that the entrepreneur chooses $\underline{q}$ and θ^* to maximize¹⁴

$$(18) \quad \Pi = p(\theta^*, \underline{q}, A_H(\theta^*, \underline{q}), A_L(\theta^*)) \\ \times (H(\bar{\theta}) - H(\theta^*)).$$

¹⁴ The choice of θ^* does not require the entrepreneur to observe type. The entrepreneur simply sets a price, and the agents themselves choose whether or not to join.

The profit-maximizing choice of $\underline{q}$ satisfies

$$(19) \quad U_a(\theta^*, \underline{q}, A_H) \\ - U_A(\theta^*, \underline{q}, A_H) \frac{\partial A_H}{\partial \underline{q}} = 0.$$

The entrepreneur chooses $\underline{q}$ to maximize the willingness to pay of the marginal member of the high group. Comparing (19) and (9) shows $\partial W_H / \partial \underline{q} > \partial \Pi / \partial \underline{q}$, which implies that the entrepreneur sets a lower $\underline{q}$ for every θ^* . Since (9) also characterizes the efficient regulation, we may conclude that regulation under profit maximization is inefficiently lax.

The profit-maximizing choice of θ^* satisfies

$$(20) \quad p(\theta^*, \underline{q}, A_H(\theta^*, \underline{q}), A_L(\theta^*)) \\ = \frac{dp}{d\theta^*} \frac{(H(\bar{\theta}) - H(\theta^*))}{h(\theta^*)} > 0.$$

Since the profit-maximizing price is positive, comparing (20) and (12) shows that there is less secession in the entrepreneurial equilibrium than in the welfare-maximizing equilibrium, holding $\underline{q}$ fixed at the high-group welfare-maximizing level. However, because the entrepreneur sets a lower $\underline{q}$ for any θ^* , the size of the secessionist group may be larger or smaller *mutatis mutandis*. Furthermore, profit maximization may lead to more or less than the efficient amount of secession, depending on whether the secession externality in (17) is larger or smaller than the market power term in (20).

The third column of Table 1 describes the entrepreneurial equilibrium in our example. This shows that the profit-maximizing value of $\theta^* = 0.807$, so roughly 20 percent of the population secedes to a profit-maximizing high group. The group qualities under profit maximization are $A_L = 0.405$ and $A_H = 3.982$. As noted above, due to the lax regulation under profit maximization, high-group quality is lower in the entrepreneurial equilibrium than in the original equilibrium or the efficient allocation.

This analysis has implicitly supposed there to be a single entrepreneur. It may be more likely that there is competition among active and potential entrepreneurs. While a complete analysis

of competition is beyond the scope of this paper, it is worth speculating on how competition could change our conclusions. One way to incorporate competition would be to suppose that there was still only one active entrepreneur, but that potential competition forced the price to zero. If the entrepreneur continued to set q according to (19), competition would involve more secession than either monopoly or welfare maximization because regulation would be less stringent. If potential competition also forced the entrepreneur to maximize welfare by setting q according to (9), then there would be no difference between competition and welfare maximization. Finally, it is possible that competition would involve many active entrepreneurs and segmentation according to type. In this case, as long as regulation was weak or there was a passive remainder, there would still be a secession externality, and inefficiency would persist.

B. Voting Equilibrium

Although single-issue elections concerning regulation are rare, there are many instances of secession where voting plays a part. Homeowner associations, for example, are typically governed by a board of directors elected by and from the neighborhood's residents. It is interesting, therefore, to consider how the equilibrium would differ if q were determined by voting in a single-issue election.

To consider voting over regulation in this model, we define $q^{MP}(\theta, A_H)$ as the most preferred level of regulation for a member of type θ . Assuming that preferences are single-peaked, $q^{MP}(\theta, A_H)$ is implicitly defined by

$$(21) \quad U_a + U_A \frac{\partial A_H}{\partial q} = 0.$$

Clearly, $q^{MP}(\theta, A_H)$ is increasing in θ . When voting over regulations, each agent is aware that any regulation will constrain other consumers and so improve group quality, implying $q^{MP}(\theta, A_H) > a(\theta, A_H)$. The policy preferred by the median voter will be able to defeat any alternative in a pairwise election. Consequently, the equilibrium value of q will be the level preferred by the median

high-group member $q^{MP}(\Theta_M, A_H)$, where the median type of a high-group member satisfies $H(\Theta_M) - H(\theta^*) = 1 - H(\Theta_M)$.

In general, there are both similarities and differences between the voting and welfare-maximizing solutions. In both cases, the level of secession is inefficient, and regulation may be a useful second-best instrument to control it. However, under voting the level of regulation is no longer efficient, given the membership of the high group. As is well known, there may be too much or too little regulation, depending on the distribution of agents and their preferences. This inefficiency has a surprising effect. If the median voter prefers inefficiently much regulation, then this situation may actually be more efficient than the situation with welfare maximization. This is of course because the inefficiency associated with voting operates in the opposite direction as the inefficiency associated with secession. Numerical solutions under voting are reported in Table 1.¹⁵

V. Conclusion

We conclude the paper by considering the implications of our analysis for various policies that have been suggested to improve the provision of local public goods like education and neighborhood quality. All of these policies relate to the failure of privatization in a setting where there is a remainder of agents who are harmed by secession.

We begin by considering school choice and educational vouchers. There are both extreme and moderate ways to implement choice in education, but most involve parents being given alternatives both to and within the traditional public sector. This is supposed to encourage innovation and responsiveness to the needs and abilities of children. In the context of our model, choice might be expressed as allowing a household to decide between the traditional public sector and a restrictive charter school, while vouchers might amount to a subsidy of secession. As

¹⁵ The voting solution is the same as the welfare-maximization solution in this example because of the separability of preferences and the uniform distribution of tastes.

noted in Section III, some secession of this kind may be efficient when agents differ in their valuation of the public service. Proposition 5 shows, however, that there will be excessive secession in this situation. Thus, educational reforms that encourage secession might have unfortunate consequences for efficiency, to say nothing of equity.

It is important to recognize, however, that our model ignores the possible productivity advantages of institutional innovations in education. For instance, charter schools might provide better education at lower cost than traditional schools. It may be worth paying the costs of secession to realize these efficiencies. Furthermore, it may be that the current system of education already involves significant secession, with some children excluded from high-quality schools by a combination of high property prices and closed enrollment policies. To the extent that this is true, enhancing choice by requiring open enrollment would make secession more difficult to accomplish, and would therefore enhance efficiency.

As noted in the introduction, private communities of various kinds have proliferated in recent years, and so has the controversy surrounding them. Propositions 2 and 3 show that this kind of secession will lead to a lower level of amenities and lower welfare for those who do not secede. Our model also suggests that the notorious regulations enacted by homeowner associations are not as bad as they might appear. In fact, an efficiency-minded planner could improve resource allocation by requiring even stronger regulations. This seemingly perverse results derives from the planner's desire to control secession by controlling regulation.

APPENDIX

PROOF OF PROPOSITION 1:

Using the envelope theorem,

$$\begin{aligned}
 (A1) \quad \partial p(\theta, \underline{q}, A_H, A_L) / \partial \theta \\
 &= U_{\theta}(\theta, \max\{\underline{q}, a(\theta, A_H)\}, A_H) \\
 &\quad - U_{\theta}(\theta, a(\theta, A_L), A_L).
 \end{aligned}$$

By the mean-value theorem for functions of several variables,

$$\begin{aligned}
 (A2) \quad \partial p(\theta, \underline{q}, A_H, A_L) / \partial \theta \\
 &= U_{\theta a}(\theta, a^+, A^+) \\
 &\quad \cdot (\max\{\underline{q}, a(\theta, A_H)\} - a(\theta, A_L)) \\
 &\quad + U_{\theta A}(\theta, a^+, A^+) \cdot (A_H - A_L),
 \end{aligned}$$

for some a^+ and A^+ on the line connecting $(A_L, a(\theta, A_L))$ and $(A_H, \max\{\underline{q}, a(\theta, A_H)\})$. If individual action and group quality are complements ($a_A > 0$) or are unrelated ($a_A = 0$), then $\max\{a(\theta, A_H), \underline{q}\} - a(\theta, A_L) \geq 0$. Since $U_{\theta a} > 0$ and $U_{\theta A} > 0$ by assumption, we have $\partial p / \partial \theta > 0$. This implies $p(\theta, \underline{q}) < 0$ for all $\theta < \theta^*$ and $p(\theta, \underline{q}) > 0$ for all $\theta > \theta^*$, which establishes the result.

If individual action and group quality are substitutes ($a_A < 0$), the argument is somewhat more involved. Suppose $\underline{q} < a(\theta^*, A_L)$. In this case, action $a(\theta^*, A_L)$ satisfies the requirement for admission to the high group. Then a type θ^* agent who joined could enjoy higher utility as a member: $U(\theta^*, a(\theta^*, A_L), A_H) > U(\theta^*, a(\theta^*, A_L), A_L)$. This contradicts $p(\theta^*, \underline{q}, A_H, A_L) = 0$. Therefore, $\underline{q} > a(\theta^*, A_L)$. Since $a_{\theta} > 0$, this implies that $\underline{q} > a(\theta, A_L)$ for all $\theta \leq \theta^*$. By (A2), this in turn implies that $\partial p(\theta, \underline{q}, A_H, A_L) / \partial \theta > 0$ for all $\theta < \theta^*$, which establishes that all agents with $\theta < \theta^*$ would choose not to join.

For $\theta > \theta^*$, if $a(\theta, A_L) \geq \max\{\underline{q}, a(\theta, A_H)\}$, joining the high group and setting $a = a(\theta, A_L)$ would give higher utility than not joining, since $U(\theta, a(\theta, A_L), A_H) - U(\theta, a(\theta, A_L), A_L) > 0$. If $a(\theta, A_L) < \max\{\underline{q}, a(\theta, A_H)\}$, $p(\theta, \underline{q}, A_H, A_L)$ is increasing in θ by (A2). Together these imply that an agent's utility is higher as a member for all $\theta > \theta^*$.

PROOF OF PROPOSITION 2:

Equations (4), (5), and (9) define $A_L(\theta^*)$, $A_H(\theta^*, \underline{q}(\theta^*))$, and $\underline{q}(\theta^*)$ as continuous functions of θ^* . Since $P(\theta^*) \equiv p(\theta^*, \underline{q}(\theta^*), A_H(\theta^*, \underline{q}(\theta^*)), A_L(\theta^*))$ is the composition of continuous functions, it is continuous itself.

By hypothesis, $P(\underline{\theta}) < 0$, and the lowest-type agents choose not to secede in equilib-

rium. If $\theta^* = \bar{\theta}$, regulation must be strong, and we have $P(\bar{\theta}) = U(\bar{\theta}, \underline{a}(\bar{\theta}), A_H) - U(\bar{\theta}, \underline{a}(\bar{\theta}), A_L)$. If we arbitrarily set $\underline{a}(\bar{\theta}) = \underline{a}(\bar{\theta}, A_H)$, then $p(\bar{\theta}, \underline{a}(\bar{\theta}, A_H), A_H, A_L) = U(\bar{\theta}, \underline{a}(\bar{\theta}, A_H), A_H) - U(\bar{\theta}, \underline{a}(\bar{\theta}, A_L), A_L) > 0$. However, since $\underline{a}(\bar{\theta})$ maximizes $U(\bar{\theta}, \underline{a}(\bar{\theta}), A_H)$, it follows that $P(\bar{\theta}) > p(\bar{\theta}, \underline{a}(\bar{\theta}, A_H), A_H, A_L) > 0$. Thus, the highest-type agents always secede in equilibrium. Note that in this case $A_H = \underline{a}(\bar{\theta})$ and $A_L = A_0$, where A_0 is defined by (13).

Since $P(\theta^*)$ is continuous, $P(\theta) < 0$, and $P(\bar{\theta}) > 0$, $P(\theta^*) = 0$ must have at least one stable solution on the interior of $[\underline{\theta}, \bar{\theta}]$.

Discussion.—It is possible that $P(\underline{\theta}) > 0$. In this case the level of regulation that would be chosen by the entire group is insufficient to deter the secession of the lowest-type agents. We ignore this case since it is neither particularly interesting nor of obvious empirical relevance.

PROOF OF PROPOSITION 3:

Equations (4) and (5) give the mappings that define A_L and A_H . Ignoring for the moment the impact of regulation, the mappings relate to each other as follows: $A_0 = H(\theta^*)A_L + (1 - H(\theta^*))A_H$, and so $A_L \leq \underline{A}_0 \leq A_H$. At an interior equilibrium, $\theta^* \in (\underline{\theta}, \bar{\theta})$, and the inequalities are strict. Further, since at least some members of the high group are regulated in equilibrium, the equilibrium value of A_H exceeds the value given in (5).

PROOF OF PROPOSITION 4:

Let $\Delta U_L = U(\theta, \underline{a}(\theta, A_L), A_L) - U(\theta, \underline{a}(\theta, A_0), A_0)$ represent the change in the utility of a

low-group member. $\Delta U_L < U(\theta, \underline{a}(\theta, A_L), A_L) - U(\theta, \underline{a}(\theta, A_0), A_0)$ since $\underline{a}(\theta, A_0)$ is utility maximizing when group quality equals A_0 . $U_A > 0$ and $A_L < A_0$ (Proposition 3) imply $\Delta U_L < 0$.

PROOF OF PROPOSITION 5:

The difference between the constrained-efficient allocation and the equilibrium is easy to see. The left side of (17) is the willingness to pay of the marginal member. It equals zero in equilibrium by (12). The right side of (17) is positive. Thus, the equilibrium involves more secession than the constrained-efficient allocation.

PROOF OF PROPOSITION 6:

Letting λ be the multiplier on the equal utility constraint and using (12), the first-order condition for θ^* implies

$$\begin{aligned} (A3) \quad & \int_{\underline{\theta}}^{\theta^*} U_A(\theta, \underline{a}(\theta^*, A_L), A_L) \frac{\partial A_L}{\partial \theta^*} h(\theta) d\theta \\ & + \int_{\theta^*}^{\bar{\theta}} U_A(\theta, \text{Max}\{\underline{a}, \underline{a}(\theta, A_H)\}, A_H) \\ & \times \frac{\partial A_H}{\partial \theta^*} h(\theta) d\theta + \lambda \frac{dp}{d\theta^*} = 0, \end{aligned}$$

so

$$(A4) \quad \lambda = - \frac{\int_{\underline{\theta}}^{\theta^*} U_A(\theta, \underline{a}(\theta^*, A_L), A_L) \frac{\partial A_L}{\partial \theta^*} h(\theta) d\theta + \int_{\theta^*}^{\bar{\theta}} U_A(\theta, \text{Max}\{\underline{a}, \underline{a}(\theta, A_H)\}, A_H) \frac{\partial A_H}{\partial \theta^*} h(\theta) d\theta}{\frac{dp}{d\theta^*}}.$$

By stability, $dp(\theta^*, \underline{q}, A_H(\theta^*, \underline{q}), A_L(\theta^*)) / d\theta^* > 0$, implying $\lambda < 0$. Using (9), the first-order condition for $\underline{q}$ is

$$(A5) \quad \frac{\partial W_H}{\partial \underline{q}} = -\lambda \left[U_a(\theta^*, \underline{q}, A_H) + U_A(\theta^*, \underline{q}, A_H) \frac{\partial A_H}{\partial \underline{q}} \right] < 0.$$

Comparing (A5) and (9) shows that the planner sets $\underline{q}$ beyond the level that maximizes the welfare of the high group.

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